1.4 Dilations and Partitions

Lesson Introduction

Benchmark Focus	MA.912.GR.2.2 Identify transformations that do or do not preserve distance.	Learning Targets	• I can dilate a line segment. • I can define midpoint using dilation and understand partitions. • I can relate scale factor to dilation.

Benchmark Coherence	Building On	Working Toward
	MA.8.GR.2.2 Given a preimage and image generated by a single dilation, identify the scale factor that describes the relationship.	N/A

Essential Questions

- How can knowing a scale factor between two figures help determine a segment length?
- How is a scale factor related to a dilation?
- How is dilation used to partition a segment?

Lesson Narrative

Students explore situations to determine how many smaller segments need to be copied to create a larger segment. Relating the principle of scale factor to dilations, students will partition segments and define midpoint.

Component Summary

1.4.1 Warm-Up (~10 min.)	Floyd Norman, Animator Career Profile *(SE p. 20)*	1.4.3 Guided Practice (15 min.)	Applying scale factor to determine length *(SE p. 24)*
1.4.2 Exploration (15 min.)	Using dilation to partition a line segment *(SE p. 21)*	1.4.4 Bring It Together (5 min.)	Using scale factor to define midpoint *(SE p. 26)*
1.4.5 Wrap-Up (~5 min.)	Reflection on whether dilation is an isometry *(SE p. 27)*		

Resources

Glossary	Materials	Additional Preparation
<u>Key Terms:</u> • Dilation • Partition • Scale Factor • Midpoint <u>Additional Terms:</u> • Isometry	• Career Profile Animator Floyd Norman • Patty Paper • Calculator	• N/A

<table>
<tr><td rowspan="4">Teacher Tips</td><td>Common Misconceptions</td><td>Things to Consider</td></tr>
<tr><td>Students may not remember the Pythagorean theorem.</td><td>Consider learning more about Floyd Norman to share his accomplishments with your students.
Consider further discussing with your students how Floyd Norman shows a growth mindset. The lesson uses the idea that mistakes are opportunities to learn. Focus on how Floyd Norman tried things and was willing to fail in order to learn.</td></tr>
<tr><td>Literacy and Language Routines</td><td>Mathematical Thinking and Reasoning (MTR) Standard</td></tr>
<tr><td>Take Turns
Students will work through 1.4.2 Exploration with a partner. Students should take turns estimating and discussing the correct procedure. As students discuss, they will decide if the estimations or procedures are reasonable and follow the expected procedure for the assigned activity. Students should continue this method while working through 1.4.3 Guided Practice.</td><td>MA.K12.MTR.1.1: Participate in Effortful Learning
Students will have opportunities throughout the lesson to cultivate a growth mindset. In 1.4.1 Warm-Up, students listen to an interview with Floyd Norman where he shares how he was able to learn from mistakes and overcome obstacles. In 1.4.2 Exploration students will recognize which student had an error in their thinking and then write an explanation of the error. Throughout the lesson students help and support each other while attempting a new method.</td></tr>
<tr><td>Differentiate Instruction</td><td colspan="2">For 1.4.3 Guided Practice, consider providing an alternate grid and changing the length of $\overline{MS}$ to 15 units. This will simplify the scale factor and keep that content focused on dilations and partitions, not fractions and decimals.</td></tr>
<tr><td colspan="3">Lesson Notes:</td></tr>
</table>

1.4.1 Warm-Up: Floyd Norman

Component Overview: *Play the Career Profile video on Disney Animator Floyd Norman for the entire class. After students watch the video (3:20 minutes), prompt them to reflect on the interview by answering the three questions independently.*

1.4.1 Warm-Up: Floyd Norman

Floyd Norman was named a Disney Legend in 2007. In 1957, Norman was employed as an inbetweener on Sleeping Beauty. An inbetweener is an animator who draws the intermediate frames between two key frames. The inbetweener's drawings create the illusion of motion.

When asked, "What is the key to growing as an artist?" Norman said, "As a student, I think the most important thing is having a willingness to learn."

1. On a scale of 1 to 10, how willing are you to learn?
   Answers will vary. I am a 10, ready to learn at any opportunity.

When asked about his career Norman stated, "Making mistakes is how we grow, how we learn." He said he had no confidence, but at least he made the first step.

2. Describe a time in your life when you had no confidence, but you made the first step and learned from making mistakes.
   Answers will vary. I had no confidence when I tried out for the soccer team. I did try out anyway and learned what I needed to do to make the team the next year.

Norman described a time when Walt Disney asked him to write. Norman made these two statements as he reflected on doing something he did not think he could do.

- "The only obstacle you face is yourself."
- "If you are willing to do the work, make that commitment, then I don't believe there is an obstacle."

3. What subject(s) are obstacles for you?
   Answers will vary. The subject that is hardest for me is Chemistry.

Floyd Norman is considered a Disney Legend. During the interview he shares how he wasn't always confident but he tried something anyway. Students may have fears about learning Geometry. During this unit help students build a culture of growth mindset.

1.4.2 Exploration: Dilating a Line Segment

Component Overview: *Students will determine how many segments can be copied to make a segment of another length and how this is related to dilations. Using scale factors, students will partition lines to find lengths.*

1.4.2 Exploration: Dilating a Line Segment

1. The diagram shows points W, W', and W''.

a. What do you notice about the points?
Answers will vary. They all are in a line and have the marks ' and ".

b. Estimate the number of times the distance between W and W' will fit between W' and W''.
Answers will vary. I estimate about three or four W to W' will fit between W' to W".

c. Two students described the relationship $\overline{WW''}$ has with $\overline{WW'}$.

Jaquaze	Andrés
$\overline{WW''}$ is three times as long as $\overline{WW'}$.	$\overline{WW''}$ is four times as long as $\overline{WW'}$.

Discuss with your partner which student is correct and the mistake the other student made. Write a brief note to the student who made the mistake.

Answers will vary.
Jaquaze, it seems you found that three segments the length of WW' would fit between W and W". That means WW" is four times as long as WW' because the total distance is four times as long.

Means
I
Start
To
Acquire
Knowledge
Experience
Skills

1.4.2 Exploration is scaffolded to allow students to engage with it with their partner rather than being guided through by the teacher. There will be an opportunity to summarize and synthesize what is discovered in the next components. Students may revise their thinking throughout the exploration.

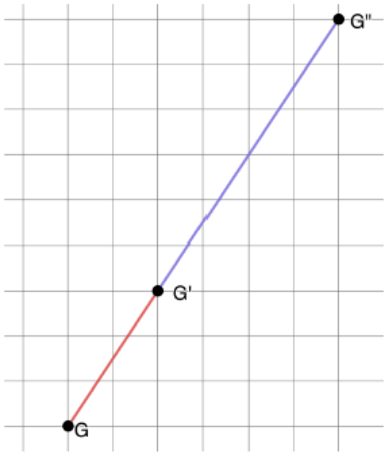
Observe and listen as students work with their partners. Make note of student ideas that you can ask them to share in the whole group 1.4.3 Guided Practice or 1.4.4 Bring It Together.

Additional scaffolding questions are provided, if needed, but this should be a primarily student-directed exploration.

If students have a hard time understanding why it is four times and not three, ask the student to try placing a number like five in for the length of $\overline{WW'}$. "If the length of $\overline{WW'}$ is 5 units, then would the length of $\overline{WW''}$ be 15 or 20 units?"

1.4.2 Exploration: Dilating a Line Segment (continued)

2. The diagram shows points G, G', and G''.



   a. Use patty paper to determine the number of copies of $\overline{GG'}$ that will fit on $\overline{GG''}$.
   Three copies of $\overline{GG'}$ will fit on $\overline{GG''}$.

   b. Discuss with your partner how to use the grid to determine the number of copies of $\overline{GG'}$ that will fit on $\overline{GG''}$. Summarize your discussion.
   Answers will vary. $\overline{GG'}$ is three squares up and two to the right. If we follow the graph up and to the right repeatedly, we will find that three $\overline{GG'}$ will fit on $\overline{GG''}$.

   c. Ask a classmate how they would use the grid to determine the number of copies of $\overline{GG'}$ that will fit on $\overline{GG''}$. Record their description and write your summary of their reason. *You are the only person who should write in the first two columns of the table.* Have the person initial next to your summary stating that your description and summary are correct.
   Answers will vary.

Description	My Summary of Their Reason	Initials

Using the Take Turns Literacy and Language Routine, students should work through question 2b by taking turns with their partner. Each time they are not talking, students should be verifying their partner's strategy by following their directions.

A ***dilation*** is a proportional increase or decrease in size in all directions.

d. Use the definition of dilation to describe why $\overline{GG''}$ is a dilation of $\overline{GG'}$.
This is a dilation because $\overline{GG"}$ is exactly three times larger than $\overline{GG'}$.

e. Fill in the blanks to complete the statements.

G' divides $\overline{GG''}$ into a ratio.

The ratio of $\frac{GG'}{GG''}$ is $\frac{1}{3}$.

Partition means to separate or to divide. A line segment can be partitioned into smaller segments. A ratio can be used to describe the partition.

f. Use the Pythagorean theorem to complete the table.

Segment	Length
GG'	$\sqrt{13}$
$G'G''$	$2\sqrt{13}$
GG''	$3\sqrt{13}$

g. Point G' is $\frac{1}{3}$ of the distance from G to G''.

h. What value can be multiplied to the length of GG' to determine the length of GG''?
3

The ***scale factor*** is the constant that is multiplied by the length of each side of a figure to produce an image that is the same shape as the original figure.

Students have had previous exposure in mathematics courses to the words, "dilation" and "scale factor," but may have not been exposed to "partition." Have the students connect their knowledge of the word partition in a non-mathematical sense prior to introducing this mathematical definition.

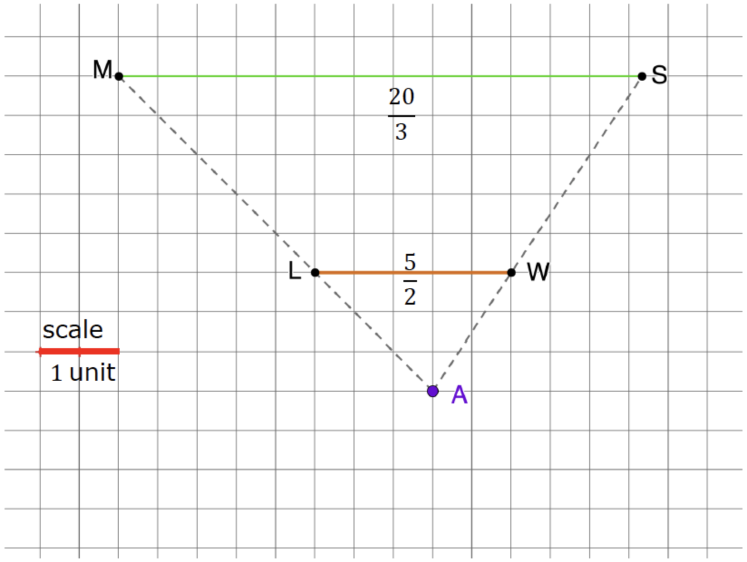
Students may provide $\sqrt{52}$ as the length of $G'G"$ and $\sqrt{117}$as the length of $GG"$. These are equivalent to the answers shown on the key and show that students know how to correctly apply the Pythagorean theorem to find distance on the coordinate plane from prior courses. Students also learned to simplify radicals in Algebra 1. Based on observation and lesson timing, determine whether this is a skill that needs to be revisited in this lesson or in the future.

1.4.3 Guided Practice: Scale Factor

Component Overview: *Students will determine the scale factor and determine all segment lengths of two overlapping triangles. They will work through questions about the length ratios to help develop the idea that scale factors are identical for all corresponding parts of similar figures.*

1.4.3 Guided Practice: Scale Factor

$\overline{LW}$ is dilated to create $\overline{MS}$. The center of dilation is point A.



1. $\overline{LW}$ has a length of $\frac{5}{2}$ units and $\overline{MS}$ has a length of $\frac{20}{3}$ units. What is the scale factor that was used to create $\overline{MS}$?
   $\frac{8}{3}$

2. Use the Pythagorean theorem to complete the table. Round values to the nearest tenth of a unit.

Segment	Length	Segment	Length
$\overline{AL}$	2.1	$\overline{AW}$	1.8
$\overline{LM}$	3.5	$\overline{WS}$	3.0
$\overline{AM}$	5.6	$\overline{AS}$	4.8

Students may struggle with fractions. This problem allows teachers to see those potential issues to determine if remediation is needed for students to interact with the types of problems expected in the course. When determining the segment lengths in question 2, students can use the fractional values or convert them to decimals, if desired. Line segment WS is given so that students can use that given value to find AS by adding the value they find for AW to WS, if needed.

Considering the dilation of a line segment in this way by looking at the relationships between the distance between the endpoints and the center of dilation helps students focus on the multiple effects of dilations. Through this problem, students explore the relationship between distance and scale factor. Allow students to share how they are solving the problem, so they are exposed to a variety of ways of thinking. For example, some students may use slope while others use the Pythagorean theorem, which allows for discussion of the relationship between the two.

3. Point L is $\frac{3}{8}$ of the distance from A to M.

4. What value can be multiplied to the length of $\overline{AL}$ to determine the length of $\overline{AM}$?
   $\frac{8}{3}$

5. Point W is $\frac{3}{8}$ of the distance from A to S.

6. What value can be multiplied to the length of $\overline{AW}$ to determine the length of $\overline{AS}$?
   $\frac{8}{3}$

7. What do you notice about each of the following ratios?
   - $LW : MS$
     $\frac{3}{8}$ is the ratio.
   - $AL : AM$
     $\frac{3}{8}$ is the ratio.
   - $AW : AS$
     $\frac{3}{8}$ is the ratio.

8. Work with your partner to explain how to find the scale factor of a dilation.
   Answers will vary. Find the ratios of each pair of corresponding side lengths. They should be equivalent. Determine the order of the ratio for the scale factor depending on which is the preimage and which is the image. For example, the scale factor from line segment LW to line segment MS is $\frac{8}{3}$, while the scale factor from $\overline{MS}$ to $\overline{LW}$ is $\frac{3}{8}$.

Students will likely confuse when to use $\frac{3}{8}$ and when to use $\frac{8}{3}$. When monitoring the students, if large numbers of students mix these up, bring the class back to whole-group discussion to clarify which is the scale factor and which is the ratio of the lengths.

1.4.4 Bring It Together: Determining Scale Factor

Component Overview: *Students decide on their favorite strategy for finding scale factors. Students then dilate a line segment using a scale factor of two to relate that to the midpoint.*

1.4.4 Bring It Together: Determining Scale Factor

As the class shares strategies for finding scale factor, choose a strategy other than yours to describe.
Answers will vary.

Strategy	I Liked the Strategy Because...

1. $\overline{XR}$ and $\overline{RK}$ are shown.

a. What is the scale factor that can be used to dilate $\overline{XR}$ to create $\overline{XK}$?
2

b. What is the scale factor that can be used to dilate $\overline{XK}$ to create $\overline{XR}$?
$\frac{1}{2}$

c. Describe point R in relation to $\overline{XK}$.
Point R is in the middle of the line. It is halfway between points X and K.

The ***midpoint*** of a line segment is the point that divides a line segment into two equal segments.

Ask different pairs of students to share their strategies. Options may include ratios of length, slope, or other methods they may develop.

1.4.5 Wrap-Up: Cool Down

Component Overview: *Students describe whether a dilation is an isometry or not.*

1.4.5 Wrap-Up: Cool Down

1. Isometry is a transformation that preserves distance and angle measure. Write a short note that describes whether a dilation preserves distance or not.
   Answers will vary. Dilation does not preserve distance. A dilation is a transformation of size, not position.

Consider providing a sentence stem for this prompt:

A dilation ________________ an isometry because an isometry preserves ___________ and a dilation ____________ preserve __________.